

CodeGraph Integration — Helix OTA

Field	Value
Revision:	1
Last modified:	2026-06-19T00:00:00Z
Status:	active

Overview

This project uses [CodeGraph](#) (@colbymchenry/codegraph) for local semantic code intelligence — a SQLite-backed knowledge graph that agents query via MCP. Per §11.4.78 (CodeGraph code-intelligence mandate), CodeGraph is the mandatory code- intelligence tool for every project under this Constitution.

Installation

CodeGraph is installed globally via npm at ~/.local/bin/codegraph:

```
npm install -g @colbymchenry/codegraph
```

Current version: 1.0.1 (codegraph --version).

The binary must be on PATH — Claude Code, Cursor, and other AGENT runtimes resolve it directly (no hardcoded host path per §11.4.78).

Configuration

Config lives at .codegraph/config.json at the project root.

Own-org submodules — INCLUDED (§11.4.79)

Submodule	Org
submodules/challenges	vasic-digital
submodules/helixqa	HelixDevelopment
submodules/ota-android-agent	HelixDevelopment
submodules/ota-artifact-validator	HelixDevelopment
submodules/ota-protocol	HelixDevelopment
submodules/ota-rollout-engine	HelixDevelopment

Submodule	Org
submodules/ota-telemetry-schema	HelixDevelopment
submodules/ota-update-engine-bridge	HelixDevelopment
containers	vasic-digital
constitution	HelixDevelopment

All own-org submodules are INCLUDED in the index so agents resolve cross-references across the entire project (§11.4.79).

Third-party submodules — EXCLUDED

Submodule	Reason
submodules/http3	Third-party, not owned

Excluded paths — rationale per §11.4.10

Pattern	Rationale
.env, .env.*	Credentials per §11.4.10
scripts/testing/secrets/**	Credentials per §11.4.10
secrets/**	Credentials per §11.4.10
*.pem, *.key	Credentials per §11.4.10
*.log	Runtime logs
qa-results/**	Raw captured evidence (gitignored per §11.4.30)
docs/qa/stress-chaos-runs/**	Stress/chaos scratch output
_exports/**, docs/**/_exports/**	Generated document exports (regenerable)
.git-backups/**	Data-safety backups (§9.2)
.codegraph/cache/**, *.db, *.db-wal, etc.	CodeGraph transient data
out/, build/, target/, bin/	Build artefacts
node_modules/	Third-party dependencies
__pycache__/, *.pyc	Python cache

Re-indexing on submodule updates

After `git submodule update --remote --merge`, re-index:

```
scripts/codegraph_setup.sh
```

Or manually:

```
codegraph sync .
codegraph status .
```

A full rebuild (cleaner, picks up engine improvements):

```
codegraph index -f .
```

MCP server wiring

CodeGraph exposes an MCP server at `codegraph serve --mcp`. The following agent configs wire it:

Claude Code

In `.claude/mcp.json` (project-scoped) or `~/.claude.json` (global):

```
{
  "mcpServers": {
    "codegraph": {
      "type": "stdio",
      "command": "codegraph",
      "args": ["serve", "--mcp"]
    }
  }
}
```

Or use the installer:

```
codegraph install --target claude --location local
```

Other agents

```
codegraph install --target cursor
codegraph install --target opencode
```

Anti-bluff verification

Run `scripts/codegraph_validate.sh` to verify the index resolves a symbol from an own-org submodule. The validation:

1. Queries CodeGraph for `Validate` in `submodules/ota-artifact-validator`
2. Asserts the query returns ≥ 1 result from that submodule
3. Reports PASS/FAIL

A FAIL means the index is stale, configured incorrectly, or an own-org submodule was accidentally excluded.

Regular updates (§11.4.80)

`scripts/codegraph_setup.sh` handles npm updates, re-indexing, and validation. Run weekly or after major submodule changes.